

Data Preparation and Tooling for Machine Learning

Author: Cyiza Landry
ETEFOP TSS

01

Answers for Question 3: File Preparation and Tooling

Data File Creation (5 Marks)

Data Format Suggestion and Justification (2 Marks)

For an image classification project like the Smart Surveillance System, you need two types of files: the image files themselves, and a file containing their labels (metadata).

File Type	Suggested Format(s)	Justification (Why it's suitable)
Image Data (The actual pictures)	JPEG or PNG	These are standard, universal formats. JPEG is good for surveillance photos (small file size, fast loading).
Label Data (The list of what's in the pictures)	CSV (Comma Separated Values)	This format is a simple table (like a spreadsheet) that perfectly links each image's file path to its correct label ("Human" or "Pet"). It is easy for Python to read.

Data File Creation (5 Marks)

File Creation Process (3 Marks)

The File Creation Process is the logical way you structure your data so the computer can easily find the images and their correct labels.

1

Directory Organization (Structure)

• Create a main folder (e.g., surveillance_data).

• Inside, create separate sub-folders for each category: Human and Pet.

• Place all the pre-processed images into their corresponding, correctly labeled folders.

2

Metadata File Generation

• Run a Python script (using the os library) that scans all these folders.

• The script creates a CSV file that acts as an index or map for the model, listing every image file path and its category.

3

Example CSV Entry

file_path, label data/Human/img_001.jpg, Human data/Pet/img_005.jpg, Pet

Dataset Import (5 Marks)

Environment Software and Importing Technique (2 Marks)

Item	Suggestion	Description
Environment Software (IDE)	Visual Studio Code (VS Code) or Anaconda (e.g., Jupyter Notebook)	These provide a powerful space to write and run your Python code.
DATASET Importing Technique	A Python function call like pd.read_csv()	This function (from the Pandas library) is used to load the organized CSV metadata file into the computer's memory, turning it into a structured table called a DataFrame.

Dataset Import (5 Marks)

Two Main Types of Dataset (1 Mark)

When preparing to train, you must split your data into two main sets to ensure you can test the model fairly:

1

Training Dataset

Used to teach the AI model (the largest portion, e.g., 70-80% of data).

2

Testing Dataset

Used only at the end to evaluate the model's final, real-world performance on data it has never seen before.

(Note: Sometimes a third set, Validation Dataset, is used during training to tune the model).





Dataset Import (5 Marks)

Role of Development Tools (2 Marks)

Development tools like Anaconda (or its components like Spyder/Jupyter) and VS Code are essential because they:

Manage Libraries

They help you install and organize all the required Python libraries (like Pandas and OpenCV) needed to read, transform, and import your data.

Execution Environment

They provide a stable environment to write, run, and debug the complex Python code required for the importing techniques and data processing without errors. They connect your code to your computer's files and memory.

Thank You